



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SHRI MASANIAMMAN AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

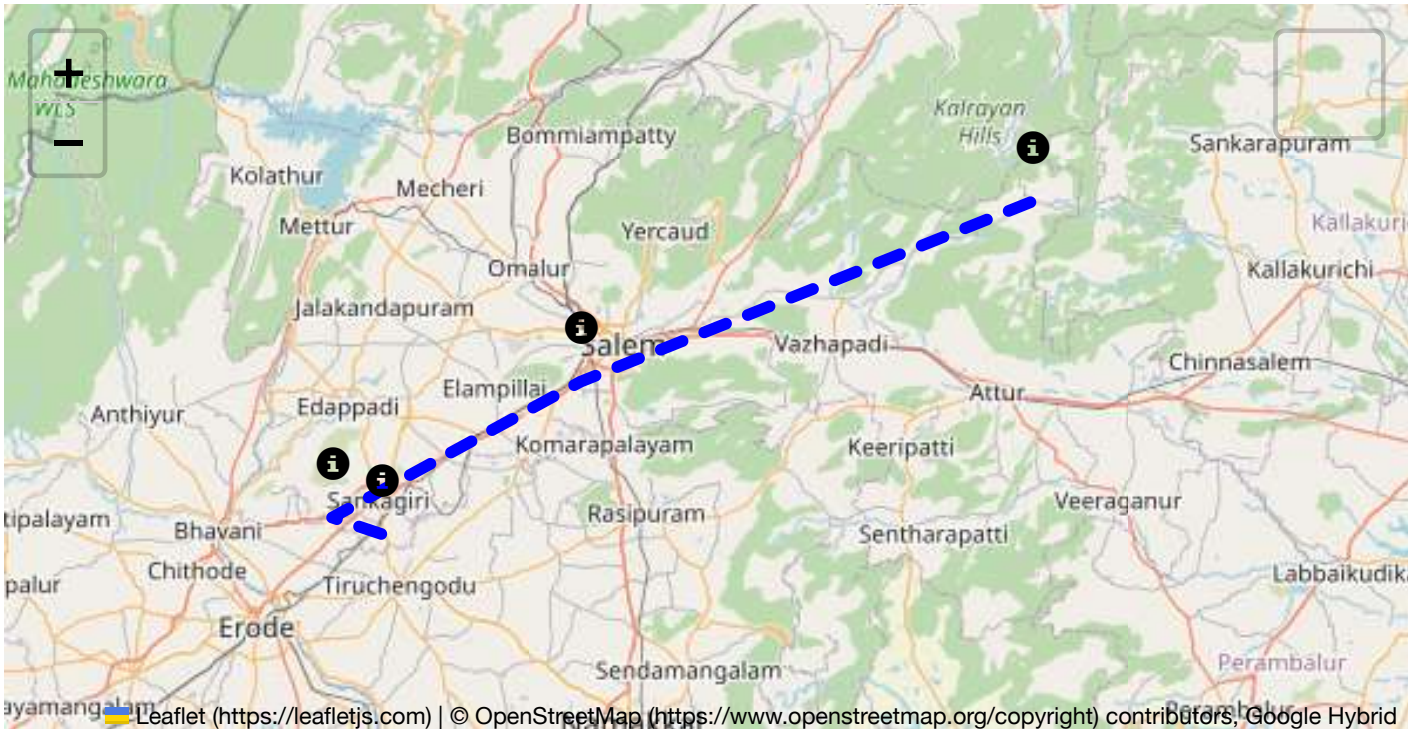
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

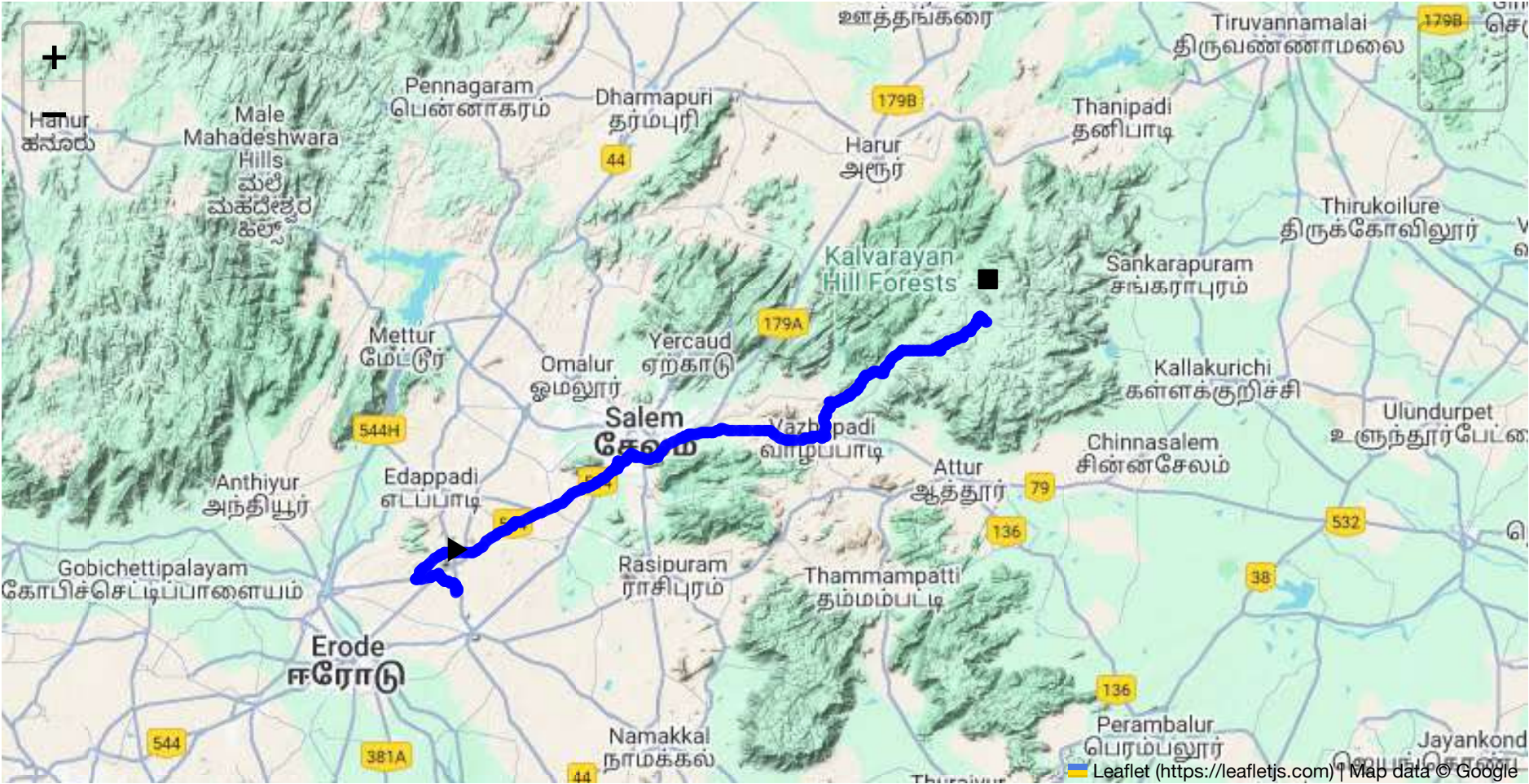
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 122.63 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours

Start: (11.4381, 77.8734)
End: (11.819747, 78.636065)



Welcome to the Journey Risk Management Study

Route Overview

The journey from Sangagiri to the Chinnakalrayan Hills is approximately 122.63 kilometers and is expected to take about 2.66 hours for heavy vehicles. The route starts from Sangagiri, passes through Padaiveedu and Uthamasolapuram, and ends at Chinnakalrayan Hills, with major roads including Bhavani Main Road and Nagapattinam - Coimbatore - Gundlupet Highway.

Weather Conditions

- Typical Weather:** The region experiences a tropical climate with hot summers, heavy monsoon rains, and mild winters. The most challenging weather occurs during the monsoon season (June to September) with potential for heavy rainfall and reduced visibility.
- Weather-Related Hazards:** Monsoon rains can cause flooding and landslides, especially in hilly areas and at river crossings. Frequent rain can create slick roads.

Traffic Patterns

- Peak Traffic Hours:** Generally, the heaviest traffic is found during morning (8 AM - 10 AM) and evening (5 PM - 8 PM) rush hours.
- Congestion-Prone Areas:** Potential bottlenecks include urban areas near Salem, especially around major intersections and highway entry and exit points.

Road Quality and Infrastructure

- General Condition:** The roads are a mix of well-paved highways and narrower local roads. The highway is generally in good condition but expect varying road quality on less traveled routes.

- **Infrastructure Concerns:** Watch for narrow bridges, sharp bends, and inadequate signage on smaller roads.

Alternative Routes

- **Emergency Alternative:** If the main route is blocked, consider taking NH44 towards Namakkal and diverting through state highways that lead back to Salem or Tamil Nadu's district roads.

Local Regulations

- **Hazardous Material Transport:** Compliance with Tamil Nadu’s rules on transportation of hazardous materials is mandatory. Follow specific guidelines on timings and permitted routes for heavy and hazardous vehicle transport.

Historical Incidents

- Records indicate occasional accidents involving heavy vehicles, often attributed to poor road conditions during rains or driver fatigue. No major incidents involving hazardous materials have been recently reported.

Environmental Considerations

- **Sensitive Areas:** The Chinnakalrayan Hills are environmentally sensitive, with protected forest areas. It’s crucial to minimize noise and pollution in these zones and adhere strictly to designated routes.

Communication Coverage

- **Coverage Analysis:** Most parts of the route have adequate mobile network coverage; however, expect potential dead zones in hilly and remote areas, particularly as you approach the Chinnakalrayan Hills.

Emergency Response Times

- **Response Times:** In urban and semi-urban areas like Salem, emergency services can respond relatively quickly (30-45 minutes). In rural or hilly areas, response could take longer (up to 2 hours), particularly if weather conditions obstruct access routes.

Overall Risk Assessment

- **Summary:** This route involves moderate risk. Main hazards include monsoon-related incidents, road quality in rural areas, and potential for miscommunication in coverage dead zones. Proper planning, adherence to regulations, and alternative route strategies help mitigate these risks. Emergency preparedness is crucial in case of weather-induced roadblocks or accidents. Regular monitoring of weather and traffic updates can enhance trip safety effectively.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63266, 78.12699	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44900, 77.87406	15 KM/Hr
4	Turn	High	11.45353, 77.85697	15 KM/Hr
5	Turn	Medium	11.45781, 77.85203	30 KM/Hr
6	Turn	High	11.46307, 77.84941	15 KM/Hr
7	Turn	High	11.45542, 77.81725	15 KM/Hr
8	Turn	High	11.45605, 77.81683	15 KM/Hr
9	Turn	Medium	11.53573, 77.95794	30 KM/Hr
10	Turn	Medium	11.62100, 78.10723	30 KM/Hr
11	Blind Spot	Blind Spot	11.62130, 78.10776	10 KM/Hr
12	Blind Spot	Blind Spot	11.61235, 78.10438	10 KM/Hr
13	Turn	Medium	11.61235, 78.10495	30 KM/Hr
14	Turn	Medium	11.66061, 78.20813	30 KM/Hr
15	Turn	Medium	11.65458, 78.37988	30 KM/Hr
16	Turn	High	11.65730, 78.40582	15 KM/Hr
17	Turn	High	11.70465, 78.41560	15 KM/Hr
18	Turn	High	11.70560, 78.41339	15 KM/Hr
19	Turn	High	11.70650, 78.41352	15 KM/Hr
20	Turn	Medium	11.70685, 78.41417	30 KM/Hr
21	Turn	Medium	11.71535, 78.43990	30 KM/Hr
22	Turn	Medium	11.72294, 78.44843	30 KM/Hr
23	Turn	Medium	11.72310, 78.44958	30 KM/Hr
24	Turn	Medium	11.74451, 78.46593	30 KM/Hr
25	Turn	Medium	11.74457, 78.46743	30 KM/Hr
26	Turn	Medium	11.74848, 78.47722	30 KM/Hr
27	Turn	High	11.74532, 78.48925	15 KM/Hr
28	Turn	Medium	11.75315, 78.49147	30 KM/Hr
29	Turn	Medium	11.75950, 78.49615	30 KM/Hr
30	Turn	Medium	11.76064, 78.49630	30 KM/Hr
31	Turn	Medium	11.77176, 78.50554	30 KM/Hr
32	Turn	High	11.77985, 78.51619	15 KM/Hr
33	Blind Spot	Blind Spot	11.78359, 78.56966	10 KM/Hr
34	Turn	High	11.78308, 78.56978	15 KM/Hr
35	Turn	Medium	11.78233, 78.56924	30 KM/Hr
36	Turn	High	11.77859, 78.56932	15 KM/Hr
37	Turn	High	11.77933, 78.57214	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
38	Turn	Medium	11.78707, 78.57397	30 KM/Hr
39	Turn	High	11.78789, 78.57500	15 KM/Hr
40	Turn	High	11.79122, 78.58364	15 KM/Hr
41	Turn	Medium	11.79106, 78.58596	30 KM/Hr
42	Blind Spot	Blind Spot	11.79265, 78.58820	10 KM/Hr
43	Blind Spot	Blind Spot	11.79217, 78.58824	10 KM/Hr
44	Turn	Medium	11.80506, 78.60908	30 KM/Hr
45	Turn	High	11.80525, 78.61498	15 KM/Hr
46	Turn	High	11.82567, 78.62743	15 KM/Hr

Risk Assessment - Elevation Changes

	Risk Type	Specific Risk	Risk Level	Coordinates	Speed Limit
0	Elevation Change	Ascent	High	11.78082960403855, 78.57217190642473	15 KMS/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
12	hospital	Thayammal Hospitals	11.6237472, 78.1177631	30 km/h	Medium
13	hospital	Vanitha Hospital	11.6239931, 78.1184622	30 km/h	Medium
15	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
16	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
17	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
18	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
19	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
20	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
21	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
22	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
23	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
24	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
29	hospital	Jaaidev Hospital	11.6655205, 78.2634736	30 km/h	Medium
31	hospital	Niharika Hospital	11.6565971, 78.3937356	30 km/h	Medium
32	hospital	Malar Hospital	11.6567854, 78.3928345	30 km/h	Medium
33	hospital	Sri Bala Hospital	11.6566667, 78.4010889	30 km/h	Medium
34	clinic	Sibi Physiotherapy Clinic	11.6561305, 78.3994263	30 km/h	Medium
35	hospital	Saraswathi Hospital, Valapady	11.6563505, 78.4009062	30 km/h	Medium
36	hospital	Sri Udaya Hospital	11.6560244, 78.4040862	30 km/h	Medium
37	hospital	Government Hospital, Valapady	11.65706, 78.403773	30 km/h	Medium
38	hospital	Devaki Hospital	11.6564189, 78.402376	30 km/h	Medium
39	hospital	Deepam Hospital	11.6576703, 78.4065155	30 km/h	Medium
40	hospital	Government hospital,Belur	11.70532, 78.419538	30 km/h	Medium
41	hospital	Government Hospital, Karumandurai	11.8227672, 78.6338324	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Sri Vidhya Bharati Matric HR Sec School	11.6186069, 78.1103239	30 km/h	Medium
14	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
25	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
26	school	Thangam ZEE public school	11.6653353, 78.251641	30 km/h	Medium
27	school	Sri Vidhya Mandir Senior Secondary School	11.668421, 78.2509573	30 km/h	Medium
28	college	AVS College od Arts and Science, Salem.	11.6642672, 78.2584134	30 km/h	Medium
30	college	Mahendra Engineering College	11.6672537, 78.2641303	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63266, 78.12699



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53573, 77.95794



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62100, 78.10723



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.62130, 78.10776



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.61235, 78.10438



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61235, 78.10495



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66061, 78.20813



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65458, 78.37988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65730, 78.40582



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70560, 78.41339



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70650, 78.41352



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70685, 78.41417



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71535, 78.43990



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72294, 78.44843



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72310, 78.44958



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74451, 78.46593



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74457, 78.46743



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74848, 78.47722



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.74532, 78.48925



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75315, 78.49147



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75950, 78.49615



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76064, 78.49630



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.77176, 78.50554



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77985, 78.51619



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.78359, 78.56966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.78308, 78.56978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78233, 78.56924



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77859, 78.56932



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77933, 78.57214



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.78707, 78.57397



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.78789, 78.57500



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.79122, 78.58364



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.79106, 78.58596



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.79265, 78.58820



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.79217, 78.58824



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.80506, 78.60908



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.80525, 78.61498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.82567, 78.62743

